Supplementary Material for An Interpretable Expert System for ERCP Cannulation Guidance: Topology-Aware Pose Coordination and Warning-Only Risk Reasoning

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This Supplementary Material accompanies the manuscript "An Interpretable Expert System for ERCP Cannulation Guidance: Topology-Aware Pose Coordination and Warning-Only Risk Reasoning." It is organized so that each section can be read without the main text. The document contains:

- Supplementary Sections S1–S5;
- Supplementary Figures S1–S14;
- Supplementary Tables S1–S4;
- Supplementary Equations (S1)–(S5);
- an independent reference list covering all sources cited in this document.

Supplementary Table S1 corresponds to the anatomical-prior validation summarized in the main text, and Supplementary Table S2 to the annotation-reliability analysis described in Section S1. Table numbers follow the migration map agreed for this submission rather than strict order of appearance within this document.

S1. Dataset Screening and Annotation Reliability

This section provides the full eligibility criteria, the extreme-scenario exclusion criteria, the key-frame construction workflow, and the annotation-reliability analysis that are summarized in the Methods section of the main text.

S1.1 Eligibility criteria and cohort assembly

Endoscopic video records of ERCP procedures performed at Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine between January 2022 and June 2024 were retrospectively screened. Inclusion required high-definition video frames captured with a therapeutic side-viewing duodenoscope that clearly demonstrated the major duodenal papilla and its surrounding anatomical landmarks, so that the anatomical spatial relationships in the retained data remained consistent with true selective biliary cannulation scenarios. Exclusion criteria were: (i) prior endoscopic sphincterotomy or papillectomy that had altered native papillary morphology; (ii) blood, bile, or debris severely obscuring the visual field such that landmarks could not be reliably identified; and (iii) surgically altered upper gastrointestinal anatomy (e.g., Billroth II or Roux-en-Y reconstruction). After eligibility screening, 257 frames were retained for catheter contour and directional vector annotation, and 398 frames for papillary landmark annotation, spanning the full cannulation phase across seven anatomical categories: catheter (contour), papillary orifice, papillary base, intestinal lumen, biliary inlet, papillary opening direction, and luminal axis.

S1.2 Key-frame construction

The dataset was assembled from retrospective video records rather than by random temporally uniform frame sampling. Key frames were selected and constructed based on the critical anatomical interaction states occurring during the cannulation process, and only frames demonstrating clear papillary structure, unambiguous catheter directionality, stable biliary inlet topology, and sufficient spatial resolution for directional reasoning were retained. All retained images underwent a standardized quality review prior to annotation to ensure anatomical continuity and spatial consistency among the papillary orifice, biliary inlet axis, catheter tip, and luminal direction. Unlike conventional dataset construction approaches oriented toward endoscopic object detection, the curation goal here was to construct an interventional navigation corpus capable of preserving genuine anatomical directional relationships, spatial topological structures, and the conditions required for subsequent biomechanical reasoning.

S1.3 Extreme-scenario exclusion criteria

Five scenario categories were excluded during quality control because visual occlusion or fundamental alteration of the

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papillary spatial geometry may limit inference by the current visual navigation framework:

- active bowel peristalsis;
- intradiverticular papilla with mucosal coverage [1];
- malignant tumor invasion of the papilla;
- active papillary hemorrhage following haemostasis;
- post-surgical anatomical reconstruction (Billroth II gastrojejunostomy or Roux-en-Y).

These categories constitute an acknowledged performance boundary of the present internal validation and are the priority targets of the planned prospective stress-test phase described in Section S4. Because such high-difficulty cases form an extremely small proportion of the source cohort, the present validation set does not contain a sample of extreme scenarios sufficient to support systematic performance evaluation under those conditions. This staged screening strategy carries explicit methodological justification: the core objective of the present study was first to validate the feasibility of stably recovering spatial directional topological relationships under standard two-dimensional endoscopic conditions, and introducing highly heterogeneous complex cases before the core mechanism had established stability would have caused anatomical variability, tissue deformation, and visual occlusion to confound simultaneously the independent evaluation of foundational geometric representation capability.

S1.4 Annotation workflow and inter-annotator agreement

All retained images underwent automated de-identification prior to annotation. A pixel-level masking algorithm detected and irreversibly obscured all text overlays in the video output containing patient identifiers, examination dates, and institutional metadata. De-identified images were stored on encrypted, access-controlled institutional servers, and data access was restricted to investigators authorized under the approved protocol.

Landmark annotations were produced by two annotators, and inter-annotator agreement was evaluated on 43 overlapping images independently labeled by both. Because all three keypoint classes (papillary orifice, intestinal lumen, and root of the papilla) were present in every image for both annotators, Cohen's κ was not applicable; agreement was therefore quantified using the intraclass correlation coefficient ICC(2,1) for the x- and y-coordinates separately, together with the paired Euclidean distance between the two annotators' points. Table S2 reports the complete agreement results. For the papillary orifice, the primary navigation landmark, agreement was excellent (ICC(2,1) = 0.949 and 0.987 for x- and y-coordinates; median paired distance 20.4 px, $\sim$0.41 mm). Agreement was lower for the intestinal lumen (ICC(2,1) = 0.197 and 0.000; median distance 311.6 px), reflecting inherent ambiguity in center-point placement within a large, deformable anatomical space, and fair-to-moderate for the root of the papilla (ICC(2,1) = 0.471 and 0.300; median distance 122.9 px), consistent with known inter-operator variability in papillary base boundary definition. Because the primary navigation task depends on the papillary orifice and its spatial relationship to the catheter tip, the lower agreement observed for auxiliary landmarks does not compromise the reliability of the navigation framework.

S2. Detailed Architecture and Feature Mapping

As discussed in the Methods section of the main text, this section provides the granular details of the task-specific pose-estimation backbones, the auxiliary depth-related feature pathway, and the complete feature mapping performed by the random forest regressor. The full network architecture and information flow, which are omitted from the main text for space reasons, are shown in Fig. S1.

A. Task-specific topology-aware pose networks

Each input frame was resized to 640×640 pixels and processed by three independent YOLOv11s-Pose networks initialized from the same pre-trained weights [2]. The catheter model used one class with 15 ordered keypoints to represent the flexible catheter contour and distal tip. The papilla model used three classes with one keypoint per class to localize the papillary orifice, papillary base, and luminal center. The directional-vector model used three classes with two ordered keypoints per class to encode the biliary inlet axis, papillary opening direction, and luminal axis. This decomposition prevents point landmarks, dense contours, and directional axes from competing within a single feature hierarchy, and it reformulates the conventional two-dimensional region-detection paradigm as an ordered keypoint vector representation.

For an anatomical structure s, the predicted topology-aware keypoint set is

$$P_s = \{p_{s,i} = (x_{s,i}, y_{s,i}, c_{s,i}) \mid i = 1, 2, \ldots, N_s\},$$

(reproducing Eq. (1) of the main text for independent readability), where $(x_{s,i}, y_{s,i})$ denotes the image-plane coordinate, $c_{s,i}$ is the confidence score, and N_s is the number of predefined topological nodes; for the catheter contour, $N_{catheter} = 15$. Selection of YOLOv11s-Pose as the backbone was guided by comparative benchmark performance in medical keypoint estimation tasks [3], and the keypoint-based strategy adopted here is contextually related to prior work on surgical instrument pose estimation in endoscopic settings [4], [5] and to high-resolution keypoint detection [6].

B. Auxiliary depth-related feature pathway

Auxiliary depth estimation was used only to provide trend-level spatial context. Appearance-enhanced, boundary, temporal-motion, and depth-related features were processed through the SingleModalEnhancer and MSAFusion pathway shown in Module B of Fig. S1. Depth-related outputs were not used to replace keypoint coordinates and were not treated as metric monocular depth measurements.

C. Feature construction and the random forest mapping

The random forest regressor does not attempt to resolve ill-posed inverse problems directly from raw RGB pixels. Instead, it maps a highly curated 12-dimensional geometric-kinematic

TABLE S2

INTER-ANNOTATOR AGREEMENT FOR THE THREE KEYPOINT CLASSES. AGREEMENT WAS ASSESSED ON 43 IMAGES INDEPENDENTLY ANNOTATED BY TWO RATERS USING THE INTRACLASS CORRELATION COEFFICIENT ICC(2,1) FOR EACH COORDINATE AXIS AND THE MEDIAN PAIRED EUCLIDEAN DISTANCE BETWEEN RATERS. LOWER DISTANCES INDICATE CLOSER AGREEMENT. THE PAPILLARY ORIFICE IS THE PRIMARY NAVIGATION LANDMARK; THE INTESTINAL LUMEN AND ROOT OF THE PAPILLA ARE AUXILIARY LANDMARKS. DISTANCES IN MILLIMETRES ASSUME THE NOMINAL 0.02 MM/PX SCALE.

Keypoint class	ICC(2,1) x	ICC(2,1) y	Median paired distance (px)	Median paired distance (mm)
Papillary orifice (primary)	0.949	0.987	20.4	∼0.41
Root of the papilla	0.471	0.300	122.9	∼2.46
Intestinal lumen	0.197	0.000	311.6	∼6.23

Note. $n = 43$ overlapping images. Cohen's κ was not applicable because all three keypoint classes were present in every image for both annotators. The millimetre-equivalent distances are derived from the nominal 0.02 mm/px conversion used throughout this study and are reported for scale context only; they are not independent physical measurements. Table numbered S2 per the migration map for this submission.

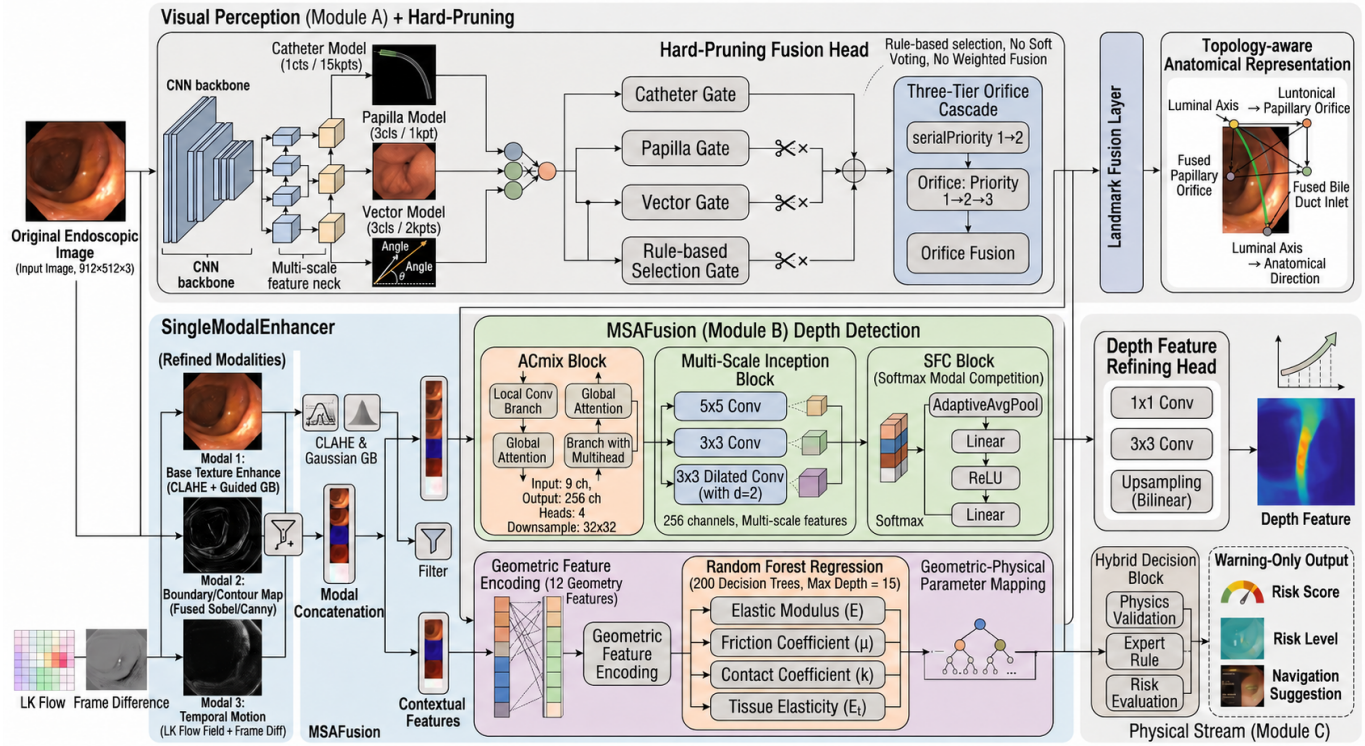


Fig. S1. **Detailed network architecture and information flow of the topology-aware physio–visual hybrid navigation framework.** Module A performs task-specific perception and anatomical coordination through three independent YOLOv11s-Pose networks: a catheter model (1 class, 15 ordered contour keypoints), a papilla model (3 classes, 1 keypoint per class), and a directional-vector model (3 classes, 2 ordered keypoints per class). Their outputs are combined by rule-based hard pruning, a three-tier orifice cascade, and an anatomical plausibility gate to construct a fused topology-aware representation of the catheter tip, papillary orifice, bile-duct inlet, and luminal axis. Module B integrates appearance-enhanced, boundary, temporal-motion, and auxiliary depth-related features through the SingleModalEnhancer and MSAFusion pathway. Module C encodes the fused anatomical geometry into geometry-derived descriptors and applies random forest regression to estimate four trend-level risk proxies, which are evaluated by a warning-only hybrid decision block that reports risk scores, risk levels, and navigation suggestions without replacing or modifying the original visual coordinates.

feature vector $\mathbf{f} \in \mathbb{R}^{12}$—which explicitly captures macro-level physical behaviors such as three-segment catheter bending angles (θ_{bend}), optical-flow-derived tip velocities (v_{motion}), and structural tissue contact intensities ($S_{contact}$)—into the continuous biomechanical property space. The model is trained to map these visual geometric features directly to four biomechanical output parameters: elastic modulus E, friction coefficient μ, stiffness ratio k, and tissue elasticity E_t. The ensemble regression is expressed as:

$$\Psi = [E, \mu, k, E_t]^T = M_{RF}(\mathbf{f}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{f}) \tag{S1}$$

where B is the total number of ensemble decision trees (set to $B = 100$ in the system experiments) and $T_b(\mathbf{f})$ denotes the output of the b-th independent regression decision tree. By interpolating between empirically calibrated rule-based tissue boundaries, the ensemble functions as a data-driven surrogate engine that transforms observable spatial deformations into mathematically bounded physical constraints, rather than act-

ing as a black-box estimator of unconstrained variables. This transition yields frame-level estimates based on NiTi wire and soft-tissue property ranges relevant to clinical ERCP, rather than a fixed stainless-steel baseline. Although this analytical formulation served as an engineering reference during early development, the deployed inference pathway relies entirely on the random forest model.

The selection of a Random Forest regressor as the biomechanical surrogate engine—rather than deep neural networks or single-variable linear estimators—was guided by three structural prerequisites. First, given the relatively constrained and specialized nature of the intraoperative dataset ($n = 45$ valid inference frames), the Random Forest architecture exhibits superior robustness against overfitting through its inherent bootstrap aggregating mechanism [7]. Second, it captures complex, highly non-linear interactions across the heterogeneous 12-dimensional feature space without requiring extensive hyperparameter tuning. Third, the ensemble decision trees entail low computational overhead, which is critical for minimizing software overlay latency and maintaining the baseline throughput required for continuous monitoring. The composite risk score is computed from Ψ and drives the alert interface illustrated in Fig. S13, which also documents the deployment-time hyperparameter settings of the perception stage (input resolution 640×640, per-class confidence thresholds, and the anatomical overlap radius D_{max} used by the cascade).

S3. Additional Coordination and Trajectory Details

This section documents the cross-model coordination layer, the complete gating algebra, and the trajectory-rendering formulation whose closed forms are omitted from the main text. The ablation evidence supporting anatomically assigned hard pruning, together with the stepwise cascade gain, is shown in Fig. S2; the full qualitative comparison across all anatomical targets is shown in Fig. S3.

A. Hard pruning and three-tier cascade

Cross-model fusion used hard pruning so that every anatomical landmark was assigned to one designated source model, and redundant predictions were discarded before downstream computation. Papillary-orifice localization followed a three-tier fallback cascade: nipple-opening direction keypoint 1 was queried first, bile-duct-inlet keypoint 1 was used when the primary source was unavailable, and the papilla-model orifice center was used as the final fallback. For the raw feature prediction vector $X = [PS_{papilla}, PS_{catheter}, PS_{ampulla}]^T$, the final valid anatomical output is defined by the composite gating mapping:

$$X_{valid} = G_3(G_2(G_1(X))) \tag{S2}$$

Here G_1 is the three-tier orifice back-off selector, G_2 the anatomical plausibility gate, and G_3 the downstream warning and trajectory-output gate. G_3 performs no coordinate averaging or correction; it preserves the selected coordinate or flags it when downstream risk thresholds are exceeded. The back-off selector G_1 is given in Eq. (4) of the main text.

B. Anatomical plausibility gate and cosine threshold

To constrain the visual stream to produce geometric solutions consistent with clinical priors, the gate incorporates the "11-o'clock principle" of selective biliary cannulation. Let the ideal biliary axis reference vector under the current two-dimensional projection coordinate system of the side-viewing duodenoscope be $v_{anatomy}$ (empirically demonstrated to typically fall near 258.36° in the image plane). The plausibility gate G_2 is then:

$$G_2(X) = \begin{cases} X, & \text{if } \frac{v_{tip} \cdot v_{anatomy}}{\|v_{tip}\|_2 \|v_{anatomy}\|_2} \geq \cos(\Delta\theta_{max}) \\ \emptyset, & \text{otherwise} \end{cases} \tag{S3}$$

where $\Delta\theta_{max}$ is the maximum deviation angle used to define the 9-o'clock to 1-o'clock prior interval in the image-plane coordinate system. If the visual prediction vector deviates outside this interval, the frame is flagged as anatomically implausible for the standard-anatomy prior, and the downstream warning logic records this plausibility flag before physical-proxy evaluation. The check cross-references the predicted bile duct–papilla angular relationship against the clinically accepted normal range and is a risk and plausibility flag rather than proof of clinical correctness.

Two clinically recognized categories of systematic failure motivate the manual override mechanism. The first is physiological variation: even in patients with entirely normal papillary anatomy, luminal rotation may shift the biliary axis such that the target direction deviates toward 9-o'clock or 1-o'clock. The second is post-surgical anatomical reconstruction: for patients with periampullary diverticulum (PAD) or juxtapapillary diverticulum (JPD) the biliary inlet direction may shift irregularly, and more critically, for patients who have undergone subtotal gastrectomy with gastrointestinal reconstruction (Billroth II gastrojejunostomy or Roux-en-Y reconstruction), the endoscopic viewing angle under the side-viewing duodenoscope undergoes fundamental reversal, inverting the biliary cannulation direction to a 5-o'clock orientation—approximately 180° from the standard prior direction. In such cases, continued enforcement of the 11-o'clock gate would generate systematic directional inversion errors, so the manual override permits the operating clinician to bypass the current prior constraint via a single-key operation.

C. Warning gate

The third-stage warning gate preserves the selected visual coordinate and only appends the downstream warning flag:

$$G_3(X) = \begin{cases} (X, 0), & \text{if } S(\Psi) = 0 \\ (X, 1), & \text{if } S(\Psi) = 1 \end{cases} \tag{S4}$$

Thus G_3 does not average, correct, or overwrite coordinates. The warning state $S(\Psi)$ itself is defined in Eq. (5) of the main text.

D. Trajectory visualization

The final fused landmark coordinates are parameterized as a smooth cubic Bézier curve through a lightweight regression

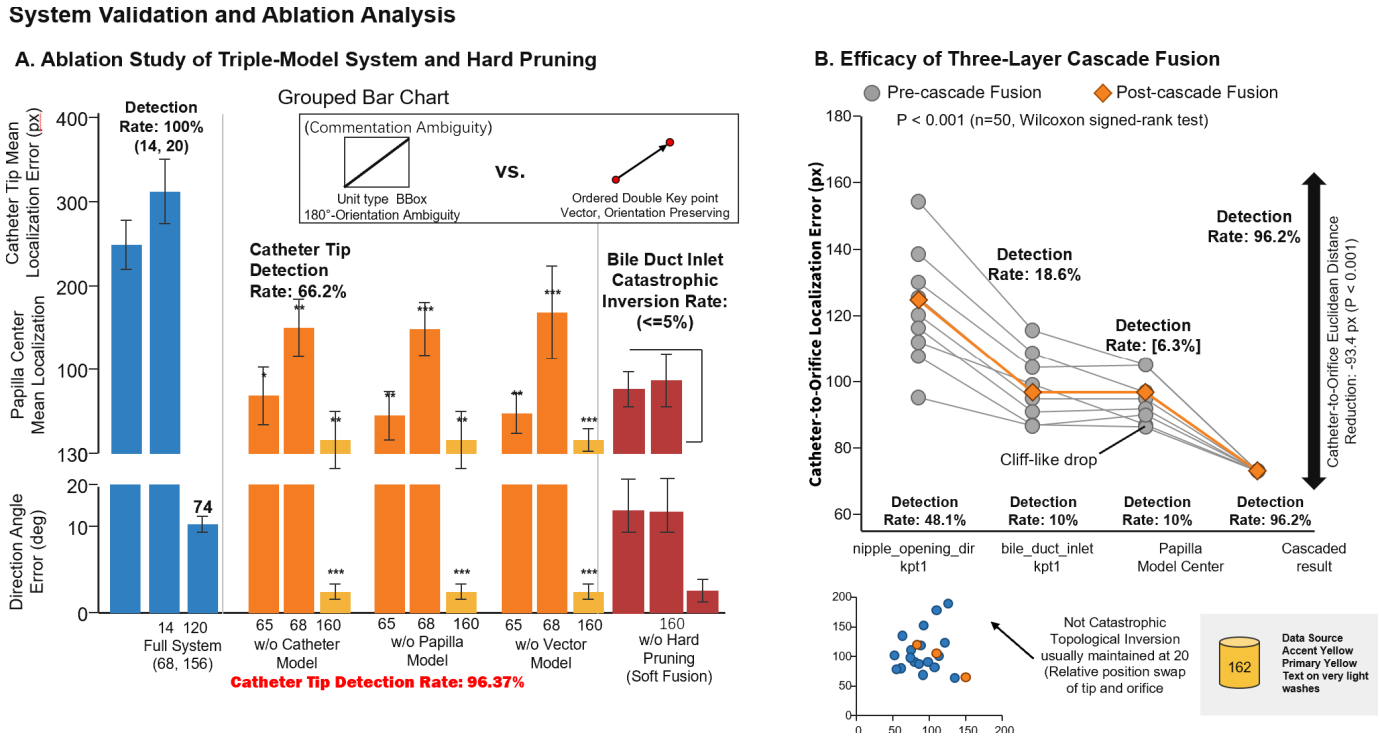


Fig. S2. **System validation and ablation analysis of the triple-model hard-pruning framework.** Panel A summarizes the ablation evidence supporting anatomically assigned hard pruning: because the luminal axis is the only structure detectable by both the papilla and vector models, unconstrained fusion would allow the less accurate source to deterministically overwrite the better one, whereas hard pruning retains a single selected coordinate per landmark. Panel B shows the three-layer cascade fusion that raises available navigation-target coordinate coverage from 48.1% (primary nipple-opening-direction source) to 90.4% (bile-duct-inlet fallback) and finally 96.2% (papilla-center fallback), while hard pruning preserves a single selected coordinate rather than averaging. Two of the 52 analyzed rows remained unfused. The cascade increases available navigation-target coordinates, not the true papillary-orifice detection rate.

head to reconstruct the predicted cannulation path:

$$\mathbf{B}(t) = (1-t)^3\mathbf{P}_0 + 3(1-t)^2 t\mathbf{P}_1 + 3(1-t)t^2\mathbf{P}_2 + t^3\mathbf{P}_3, \quad t \in [0,1] \tag{S5}$$

where $\mathbf{P}_0$, $\mathbf{P}_1$, $\mathbf{P}_2$, and $\mathbf{P}_3$ are control points used for visual trajectory rendering. This path serves only as a visual overlay for clinical display and is not fed back into coordinate control or the coordinate pipeline. An example of the rendered overlay, together with the prospective clinical-evaluation concept, is shown in Fig. S12.

E. Closed-loop correction hierarchy and the compensatory paradox

As detailed in the main text, empirical validation of the closed-loop correction hierarchy revealed that coordinate-level correction degraded localization accuracy in more than 80% of triggered interventions. The complete algebraic treatment of the resolution mismatch underlying this behavior, and the joint objective function whose minimization characterizes conventional strong-constraint closed-loop control, are given in Eq. (6) of the main text.

S4. Anatomical Prior and Future Validation

This section provides the complete analysis of the 11-o'clock anatomical prior, whose summary statistics appear in the main text, and the detailed design of the planned validation phases that the main text summarizes in a single sentence. The predicted papillary-orifice probability heatmap and the polar distribution of predicted bile-duct entry directions are shown in Fig. S4, and the corresponding validation statistics are reported in Table S1.

A. Complete analysis of the 11-o'clock prior

The foundational clinical heuristic for selective biliary cannulation holds that the common bile duct orifice is located at approximately the 11-o'clock position relative to the longitudinal axis of the papilla, a spatial prior internalized by experienced endoscopists through hundreds of supervised cases. We validated both whether this anatomical relationship is faithfully encoded in the training data and whether the trained model reproduces it.

Among all 52 ground-truth annotations, 96.15% of bile duct–papilla angular relationships fell within the clinically accepted 9-o'clock to 1-o'clock range, with a mean ground-truth direction of 258.36° in the two-dimensional endoscopic image coordinate system. In the subset of frames in which both keypoints were simultaneously detected ($n = 19$), 94.7% (95% CI: 75.4–99.1%) of predicted directions successfully reproduced this normal anatomical range, with a median angular deviation between predicted and ground-truth directions of 16.23°. This close concordance indicates that the network has internalized the underlying anatomical distribution—accommodating natural inter-patient variability—rather than

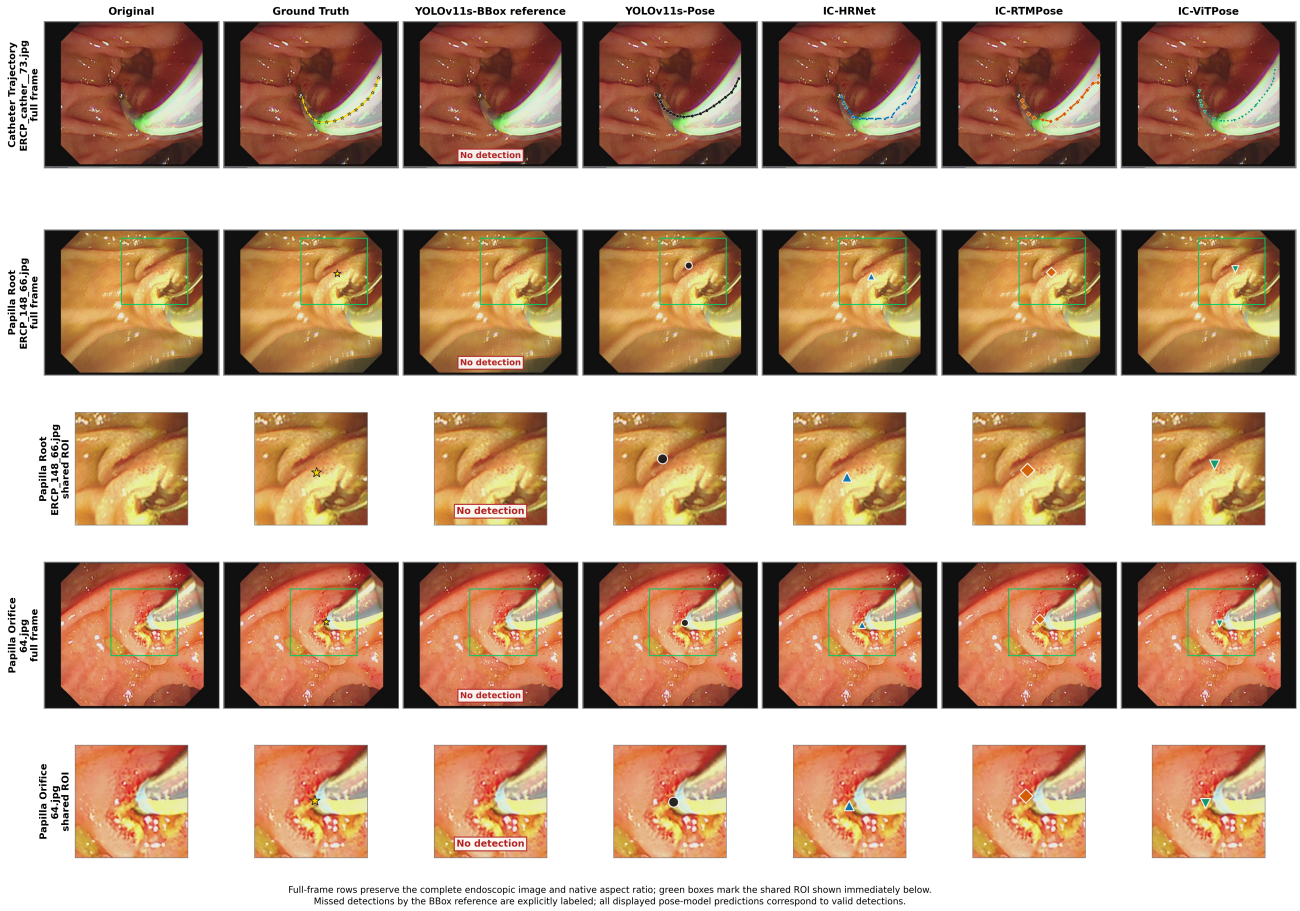


Fig. S3. **Complete qualitative comparison of pose-estimation results in representative ERCP frames.** Five row groups are shown: catheter trajectory (full frame), papillary root (full frame), papillary root (shared region of interest), papillary orifice (full frame), and papillary orifice (shared region of interest). Within each row group the columns are: the original endoscopic frame, the ground-truth annotation, a YOLOv11s-BBox reference, and predictions from YOLOv11s-Pose, IC-HRNet, IC-RTMPose, and IC-ViTPose, all displayed in identical image coordinates. The catheter example compares whole-trajectory coherence and alignment of the cannulation-facing tip, defined as the catheter end nearest the papillary orifice; the papillary-root and papillary-orifice examples illustrate model-dependent landmark dispersion around the corresponding ground-truth points. Full-frame rows preserve the complete endoscopic image and native aspect ratio; green boxes mark the shared regions of interest shown immediately below. Missed detections by the BBox reference are explicitly labeled, whereas all displayed pose-model predictions correspond to valid detections. The main text reproduces only the catheter and papillary-orifice subsets of this figure; all remaining targets are provided here.

TABLE S1

ANATOMICAL PRIOR VALIDATION (11-O'CLOCK DIRECTION, $n = 52$). ANGULAR RELATIONSHIPS BETWEEN THE BILE DUCT AND THE PAPILLA WERE COMPUTED FOR ALL 52 GROUND-TRUTH ANNOTATIONS AND FOR THE SUBSET OF FRAMES YIELDING SIMULTANEOUS DUAL-KEYPOINT PREDICTIONS. THE ACCEPTED NORMAL RANGE IS THE CLINICALLY DEFINED 9-O'CLOCK TO 1-O'CLOCK INTERVAL.

Category	Metric Description	Result	95% CI (Wilson)
Ground Truth ($n = 52$)	In 9–1 o'clock range	96.15%	–
	Mean Angle	258.36°	–
Prediction ($n = 19$)	In 9–1 o'clock range	94.74% (18/19)	75.4%–99.1%
	Median Angular Deviation	16.23°	–

Note. The mean ground-truth direction of 258.36° corresponds to the 8.6-o'clock position by raw image-plane metrics, but corresponds to the trajectory of the clinical 11-o'clock direction once the coordinate transformation and lateral inversion inherent to the side-viewing duodenoscope projection geometry are taken into account. Table numbered S1 per the migration map for this submission.

collapsing into a rigid, overfitted directional pattern. Collectively, these results serve a dual function: they validate the anatomical fidelity of the training corpus itself, and they provide empirical justification for the 11-o'clock consistency check embedded within the fusion module.

It is important to emphasize that the "11-o'clock direction" in this study does not represent a fixed and universally applicable anatomical rule, but rather reflects the commonly observed empirical spatial tendency under standard side-viewing duodenoscopy conditions. In genuine clinical settings, the biliary axis may deviate significantly due to luminal rotation, periampullary diverticulum (PAD), juxtapapillary diverticulum (JPD), local inflammatory edema, and post-surgical anatomical alterations such as Billroth II or Roux-en-Y reconstruction; in

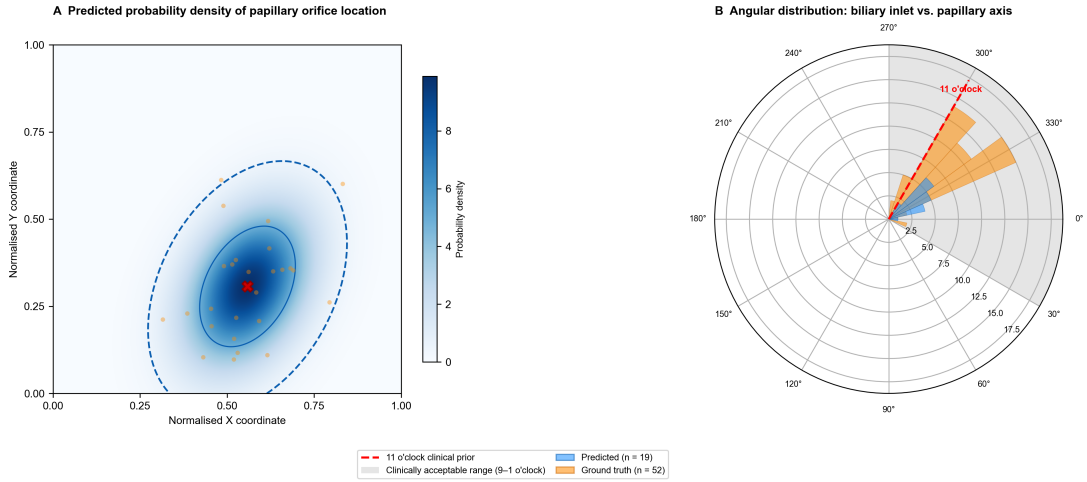


Fig. S4. **Anatomical prior learning and directional consistency.** (A) Predicted papillary-orifice probability heatmap, showing that the network localizes the papillary opening as a spatially concentrated region rather than a diffuse response. (B) Polar distribution of predicted bile-duct entry directions relative to the papillary axis. Each angular observation corresponds to a frame in which both keypoints were simultaneously detected ($n = 19$). The clinically accepted 9-o'clock to 1-o'clock normal interval is shaded; 94.74% of predicted directions fell inside this interval, and the median angular deviation from ground truth was 16.23°. Because the 11-o'clock direction is an empirical spatial tendency under standard side-viewing duodenoscopy rather than a rigid anatomical law, predictions outside the shaded interval are recorded as plausibility flags rather than treated as errors.

certain cases, the cannulation path may even approximate a reverse 5-o'clock orientation. Accordingly, the directional consistency check in this system is fundamentally an anatomical plausibility reference rather than a rigid geometric constraint.

B. Cross-center generalizability considerations

Cross-center validation will require IRB negotiation with a minimum of two to three partner institutions; based on typical ethics review timelines, a realistic target for initiating external validation is within 6–12 months of publication. Cross-center generalizability may be affected by systematic differences in endoscope model and image resolution (e.g., Olympus vs. Pentax vs. Fujifilm platforms), white-balance and auto-gain settings that alter the chromatic characteristics of the papillary surface, operator cannulation technique and experience distribution, and patient population composition including the prevalence of anatomical variants such as periampullary diverticulum.

C. Frame-rate adequacy for advisory guidance

The current pipeline operates at 4.34 FPS (∼230 ms per frame). Two senior ERCP endoscopists (each with >1,000 cases of selective biliary cannulation experience) considered this update rate compatible with an offline warning-only feasibility prototype during system design; this consultation does not constitute prospective usability or safety validation. The clinical reasoning behind this judgment reflects the specific cognitive structure of ERCP cannulation: unlike continuous closed-loop interventions such as robotic vascular catheterization, ERCP cannulation is a tactile-dominant, intermittently confirmed procedure. A further physiological basis is the relative anatomical stability of the system's primary navigational targets: papillary orientation and biliary axis direction are anatomical features that typically evolve more slowly than catheter or guidewire motion itself. Consequently, near-real-time spatial guidance in ERCP does not necessarily require the high frame rates demanded by continuously changing environments—such as cardiovascular intervention or continuous robotic guidance—where instrument trajectory and target geometry evolve simultaneously on millisecond timescales. Operators primarily rely on haptic feedback for real-time micro-adjustment, while visual guidance serves as a directional plausibility reference at discrete decision points, for example before each guidewire advancement attempt. In this context, sub-second visual updates at 4.34 FPS are operationally adequate, and the absence of continuous visual feedback between decision points does not impair procedural safety because the operator's motor loop is governed by tactile rather than visual input. The warning-only architecture provides a further structural basis for this adequacy: by design the system never injects predictions into the operator's motor loop and does not participate in trajectory control, so it solely flags risk states and preserves the underlying high-fidelity visual coordinates. This structural decoupling eliminates the "false spatial confidence" risk that motivates 15–20 FPS thresholds in autonomous navigation contexts, where delayed visual feedback may cause a closed-loop controller to act on stale spatial estimates. Comparable regulatory precedent exists: the FDA-cleared GI Genius polyp detection system operates at a similarly constrained frame rate in clinical endoscopy without compromising patient safety, demonstrating that frame-rate requirements in AI-assisted endoscopy are procedure-specific rather than universally governed by general real-time AI standards [8]. The current retrospective design nevertheless precluded collection of structured subjective usability metrics (e.g., NASA-TLX workload assessments) or direct comparisons of operator performance across different frame rates, and validating the ergonomic acceptability of this sub-second

update frequency remains a priority for the prospective live-procedure phase.

D. Planned prospective validation

Stress-test validation. The highest-priority near-term direction is prospective validation on the five anatomically challenging scenario categories excluded from the present study: periampullary diverticulum (PAD), juxtapapillary diverticulum (JPD), post-surgical reconstruction (Billroth II gastrojejunostomy and Roux-en-Y), active papillary hemorrhage following haemostasis, and intradiverticular papilla with mucosal coverage. The target acquisition goal is 50–100 cases, with case recruitment anticipated to require approximately 12 months under the existing IRB-approved protocol at Xinhua Hospital, subject to negotiation with partner institutions for post-surgical reconstruction cases, which are encountered at lower frequency in standard ERCP practice. Performance on this stress-test set will be reported against the same primary endpoints used in the present study—catastrophic directional inversion rate and navigation-target coordinate availability—to enable comparison with the standard-anatomy internal feasibility baseline established here. This phase will help assess whether the topology-aware keypoint architecture generalizes beyond standard-anatomy cases and will inform the case-mix design of the subsequent multicenter prospective trial.

Phase II: external validation. Following stress-test validation, a Phase II multicenter external validation study is planned across a minimum of three independent centers with a combined target enrollment of $\geq$500 cases, encompassing diverse equipment platforms, operator experience levels, and patient populations. This phase will constitute the primary evidence base for any future regulatory submission and will determine whether the performance gains demonstrated in the present single-center study are reproducible under real-world clinical heterogeneity.

Prospective trial endpoints. A prospective randomized-controlled trial should employ clearly defined clinical endpoints: the primary endpoint is the first-attempt selective biliary cannulation success rate; secondary endpoints are the cumulative success rate within the first three attempts, cannulation success within 5 minutes, total cannulation time, ultimate procedural success rate, and total number of cannulation attempts; safety endpoints are the post-ERCP pancreatitis rate, perforation rate, clinically significant hemorrhage rate, and post-procedural infection rate, implemented across multiple centers and operators to ensure generalizability [9]. Sample-size planning for such a trial, assuming an improvement from 40% to 60% in first-attempt success, is reported in the Methods section of the main text.

Additional development directions. Further augmentation of difficult-cannulation and complex-anatomy datasets is required to improve model robustness. The planned optimization pathway includes TensorRT FP16 quantization (target: $\sim$12 FPS), knowledge distillation from YOLOv11s to YOLOv11n (target: $\sim$18 FPS), and unified multi-head shared-backbone architectures (target: $\sim$25 FPS), with edge deployment validation on the NVIDIA Jetson Orin platform as the final translational step. Objective training assessment tools based on this framework represent a further application [10]. Integration of real-time force sensing—for example, via fiber Bragg grating (FBG)-instrumented catheters capable of measuring contact force and tissue elasticity directly at the catheter tip—would provide the ground-truth biomechanical labels required to cross-validate and recalibrate the visual geometry-driven risk index, representing a critical pathway toward upgrading the physical stream from a trend-level safety monitor to a quantitatively validated biomechanical measurement tool [11], [12]. The most clinically impactful application of the system may lie not in assisting experts with routine cannulation, but in providing reliable spatial guidance for difficult papilla scenarios—including periampullary or intradiverticular papilla, post-surgical anatomical alterations, and obscured or atypically positioned orifices—where even experienced operators face higher failure rates and the safety margin between successful deep cannulation and procedure-related complications is narrowest.

Staged validation as a methodological design choice. The staged validation strategy adopted in this study—initiating with anatomically standard cases before expanding to complex scenarios—reflects a deliberate design choice consistent with the phased validation framework recommended for diagnostic accuracy studies [13] and specifically adapted for emerging AI-assisted diagnostic devices [14], [15]. Introducing anatomically heterogeneous cases such as periampullary diverticulum, juxtapapillary diverticulum, or post-surgical reconstruction at an early stage, before the core spatial topology mechanism had demonstrated stability, would conflate representational capability with case-mix complexity and render it impossible to attribute performance differences independently to the geometric architecture rather than to anatomical variability. This staged inclusion approach mirrors the phased case-selection framework described in item 22 of the STARD 2015 reporting guideline [13] and the validation trajectory of other clinically deployed endoscopic AI systems, which have uniformly progressed from controlled single-center internal validation to multicenter prospective evaluation as the evidence base matures.

S5. Extended Robustness, Stress-Testing, and System-Level Analyses

This section collects the extended engineering analyses, robustness and stress-test evidence, and system-level documentation that support the main text but are not required for its central argument. It also contains the complete representation-comparison table whose summary is reported in the main text.

A. Direction-estimation representation comparison

Table S3 reports the complete comparison of the four direction-estimation representations evaluated on the same held-out validation dataset: the archived axis-aligned bounding-box diagonal method, a GT-BBox ROI + PCA baseline in which validation ROIs were cropped from annotated bounding-box labels before Otsu thresholding and PCA/moments orientation estimation, a derived orientation-aware YOLO-OBB baseline trained by converting the same

vector labels into oriented bounding-box labels, and the ordered keypoint vector representation. All methods were evaluated from the same held-out validation dataset; denominators differ because each method yields its own set of valid computable directional outputs under representation-specific prediction and matching requirements. This unified-dataset reporting supports comparable interpretation of method-specific directional performance without treating denominator differences as separate data sources. The GT-BBox ROI + PCA baseline uses ground-truth bounding-box ROI crops rather than detector-predicted boxes and therefore represents a traditional post-processing reference rather than a deployable detector; the derived YOLO-OBB baseline provides a model-comparison reference point. Both remained less accurate than the ordered keypoint vector representation.

B. Engineering failure-mode assessment

Although the proposed navigation framework showed stable outputs in the retrospective internal stress test, several representative engineering failure modes were observed during log review. To delineate the prototype's operational boundaries, anomalous engineering-log states were analyzed—moderate constraint violation, complete detection failure, and severe catheter bending—based on the physical constraint engine and navigation output logs. These failure modes were derived from retrospective engineering-log analysis and do not represent clinically confirmed diagnostic categories.

Among all analyzed frames, moderate constraint violations accounted for 15.4% (8/52), typically manifesting as directional constraint anomalies or increased potential tissue interaction risk; such cases trigger a yellow-to-red warning prompt advising the operator to pause and re-confirm the cannulation direction. Complete detection failure accounted for 13.5% (7/52): in these frames the system was unable to generate valid navigation outputs due to severe occlusion of critical anatomical structures or marked degradation of image quality, and it proactively output a null prediction state. Severe catheter bending accounted for 7.7% (4/52), typically manifesting as marked deviation of the catheter from the target direction and triggering a red risk alert advising the operator to adjust catheter orientation before proceeding. Table S4 summarizes these statistics. The 7/52 engineering-log failure count and the 2/52 unfused outputs of the navigation-target cascade describe different endpoints; the available summary data do not establish which frames overlap.

These engineering-log observations suggest that the primary observed failure sources in this internal dataset were associated with degradation of visual information quality and spatial constraint anomalies, rather than obvious breakdown of the topological representation itself. When critical anatomical landmarks remain visible, the topology-aware landmark representation and three-tier cascade fusion mechanism maintain stable spatial localization performance. Under conditions of severe occlusion, abnormal catheter deformation, or complex tissue interactions, however, system performance may still degrade, so the outputs should remain warning-only and require independent operator judgment. The current dataset lacks frame-level expert annotation for specific clinical-pathological states; this analysis should therefore be regarded as an engineering-level failure-mode assessment rather than a clinical diagnostic classification study. The failure-mode denominators in this section count all 52 analyzed validation frames, which is a broader denominator than the 45 valid-inference frames used for the biomechanical compliance analysis in the main text; the two figures are not directly comparable.

C. Robustness and stress-test evidence

Following standard environmental testing, common visual disturbances encountered in ERCP were simulated, including glare interference and endoscope jitter, and the system was evaluated under a retrospective engineering stress-test protocol. Within this retrospective internal setting, outputs remained available and the decision strategy remained unchanged across the simulated disturbance cases evaluated; these data do not establish performance under prospectively collected extreme conditions. Fig. S9 documents the stress-testing protocol covering glare interference and endoscope jitter. Fig. S6 provides additional plots of directional-error distributions and keypoint localization tolerance. The empirical basis of the warning-only architecture is shown in Fig. S7, which compares baseline visual error with error after geometry-derived risk-proxy intervention and illustrates the observed signal-pollution region. Fig. S8 presents the multidimensional performance envelope summarizing the balance of the physio-visual architecture across accuracy, robustness, and safety, and Fig. S10 integrates the core findings of the five experiments into a unified view. Fig. S5 documents the logical flow of the physics-sensing engine, that is, the pipeline of biomechanical feature extraction and risk evaluation whose outputs drive the warning-only decision block.

D. Regulatory context of the warning-only design

Under the FDA's Software as a Medical Device (SaMD) framework [16] and the EU Medical Device Regulation (EU MDR 2017/745), AI systems that autonomously override clinical measurements or modify operator inputs are subject to substantially higher levels of pre-market scrutiny and post-market surveillance than advisory-only systems. By confining all outputs to alert and reference functions—without intervening in coordinate data or trajectory control—the present system falls within the advisory decision-support category, reducing the regulatory burden for clinical deployment and minimizing the liability exposure associated with automated intervention errors. This positioning is consistent with the Good Machine Learning Practice principles for medical device development [15], which emphasize that the intended clinical role of an AI-enabled device should determine its risk categorization and its evidence requirements. The specific clause-level regulatory analysis supporting this categorization, together with the market-specific conformity pathways for the EU and US jurisdictions, remains a separate regulatory workstream and is outside the scope of this technical report.

TABLE S3

COMPLETE COMPARISON OF DIRECTION-ESTIMATION REPRESENTATIONS FOR ERCP ANATOMICAL AXES. MEAN ABSOLUTE ANGULAR ERROR (MAE) AND CATASTROPHIC INVERSION RATE ARE REPORTED FOR THE AXIS-ALIGNED BBOX DIAGONAL METHOD, THE GT-BBOX ROI + PCA BASELINE, THE DERIVED YOLO-OBB BASELINE, AND THE ORDERED KEYPOINT VECTOR REPRESENTATION. ALL METHODS WERE EVALUATED FROM THE SAME HELD-OUT VALIDATION DATASET ($n = 52$ IMAGES); DENOMINATORS INDICATE METHOD-SPECIFIC VALID COMPUTABLE DIRECTIONAL OUTPUTS, NOT DIFFERENT DATASETS.

Methods	MAE (°)	Inversions (>90°)	95% CI
BBox Diagonal Method	107.0	61.54% (96/156)	53.7%–68.8%
GT-BBox ROI + PCA Baseline	29.4	30.57% (48/157)	–
Derived YOLO-OBB Baseline	25.8	17.81% (26/146)	–
Ordered Keypoint Vector (Ours)	13.6	2.22% (2/90)	0.6%–7.7%

Note. MAE is the mean absolute angular error against ground-truth annotation with wrap-around correction. Catastrophic inversion is defined as an absolute angular error exceeding 90°. The BBox and keypoint denominators differ for the representation-specific reasons described above and do not establish a paired comparison.

TABLE S4

STATISTICS OF SYSTEM FAILURE MODES. CLASSIFICATION OF ANOMALOUS ENGINEERING-LOG STATES OBSERVED ACROSS THE ANALYZED VALIDATION FRAMES. FAILURES WERE IDENTIFIED FROM PHYSICAL CONSTRAINT ENGINE AND NAVIGATION OUTPUT LOGS; THEY ARE ENGINEERING-LEVEL STATES RATHER THAN CLINICALLY CONFIRMED DIAGNOSTIC CATEGORIES.

Failure type	Description	Ratio
Moderate constraint violation	Directional constraint anomaly or elevated predicted tissue interaction risk; yellow-to-red warning	15.4% (8/52)
Complete detection failure	Severe occlusion or degraded image quality; null prediction state output	13.5% (7/52)
Severe catheter bending	Marked deviation of the catheter from the target direction; red risk alert	7.7% (4/52)

Note. Denominator is the 52 analyzed validation frames. Failure categories are not mutually exclusive at the frame level in all cases; percentages are reported independently for each category. Table numbered S4 per the migration map for this submission.

E. Discussion of the compensatory paradox and its implications

The compensatory paradox documented in the main text has a direct structural interpretation that is worth stating explicitly. The random forest engine operates by translating sequential visual landmarks into macro-level physical constraints rather than pixel-level coordinate vectors. Because the spatial resolution of visual perception (Δx_{visual}) and the resolution of physical proxy derivation (Δx_{physic}) differ by an inherent order of magnitude ($\Delta x_{visual} \ll \Delta x_{physic}$), directly solving a closed-loop joint optimization equation inadvertently forces lower-resolution macro-estimates to overwrite high-resolution pixel structures, precipitating high-frequency signal contamination. This is not a deficiency of the random forest estimator itself; it is a consequence of placing a macroscopic constraint term in a position of authority over a microscopic measurement. The warning-only architecture resolves this by removing the authority relation while retaining the monitoring function, so that the physical stream can continue to flag adverse trends without being able to corrupt the geometric signal it is monitoring.

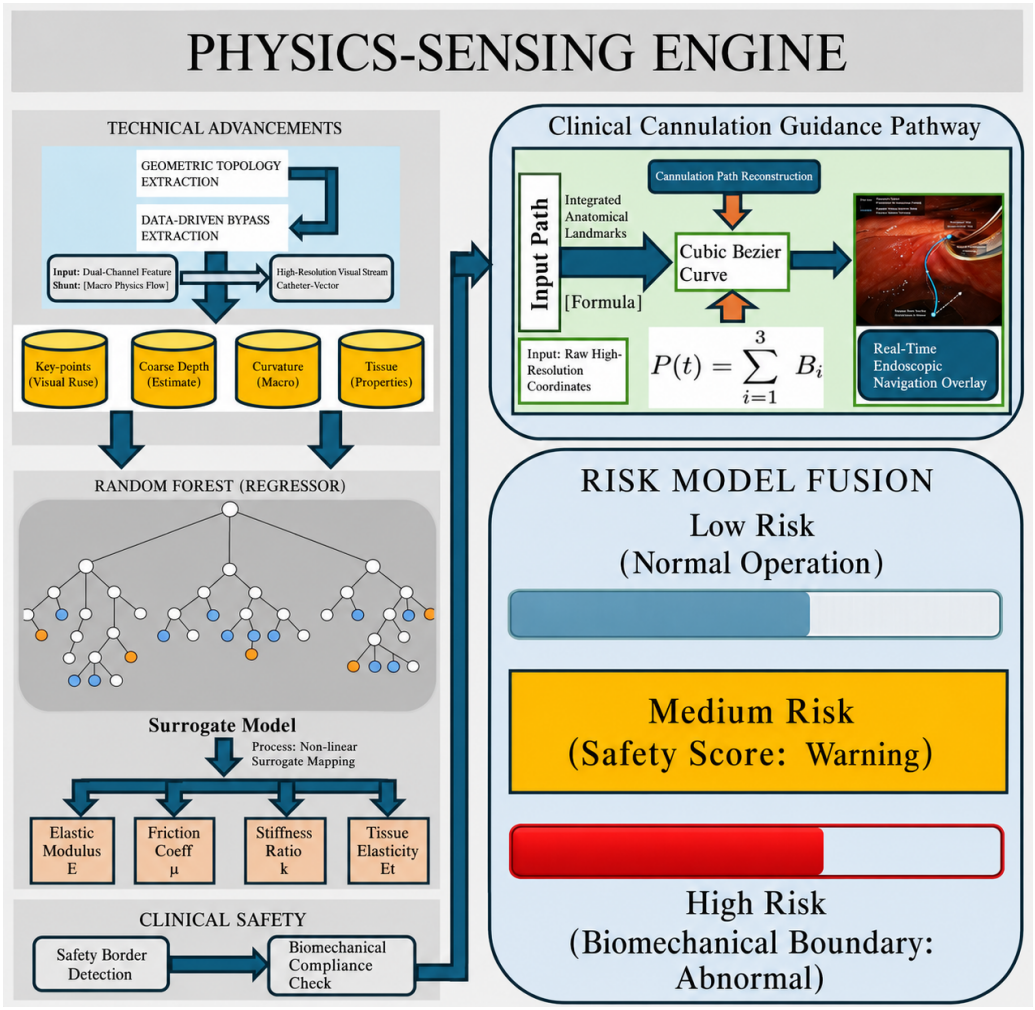


Fig. S5. **Physics-sensing engine — logical flow.** Pipeline of biomechanical feature extraction and risk evaluation. Fused visual landmarks are converted into geometric descriptors, mapped by the random forest ensemble into four trend-level risk proxies, and evaluated against biomechanical boundary conditions by a hybrid decision block. The block emits risk scores, risk levels, and navigation suggestions, and never writes back to the visual coordinate stream. This figure documents the internal structure of the module whose external behavior is summarized in the main text.

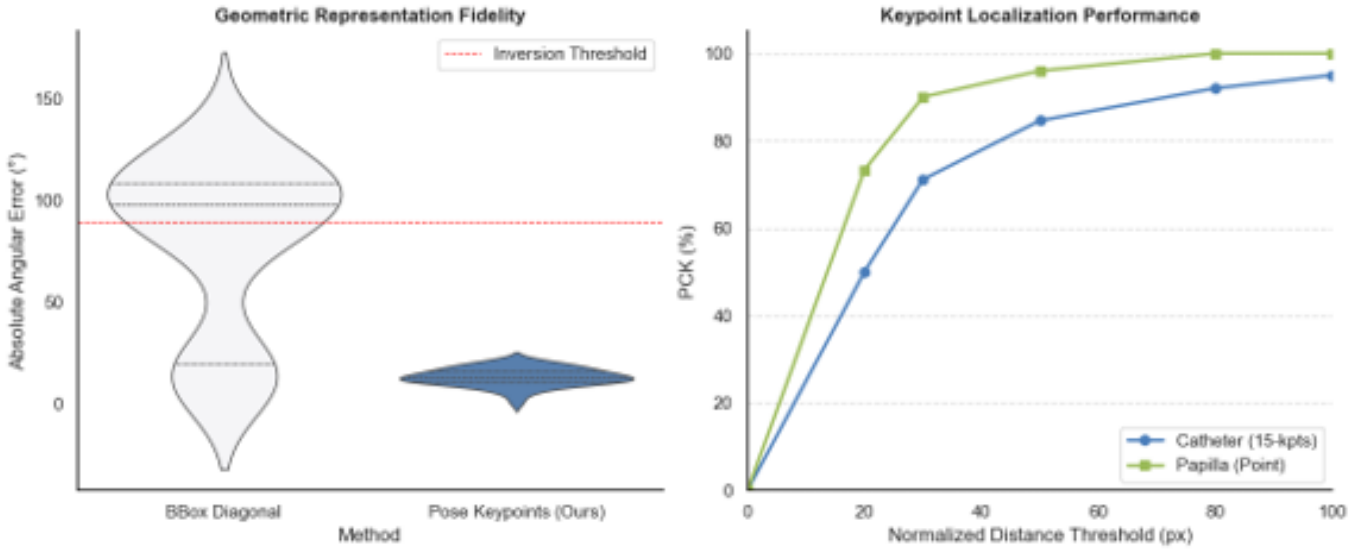


Fig. S6. **Directional representation and keypoint localization summaries.** The left panel compares absolute angular-error distributions for the BBox diagonal and ordered keypoint representations, with the catastrophic-inversion threshold marked. The right panel shows catheter and papilla PCK as the pixel-distance tolerance increases. These plots summarize the internal validation results and do not measure low-light or specular-reflection robustness.

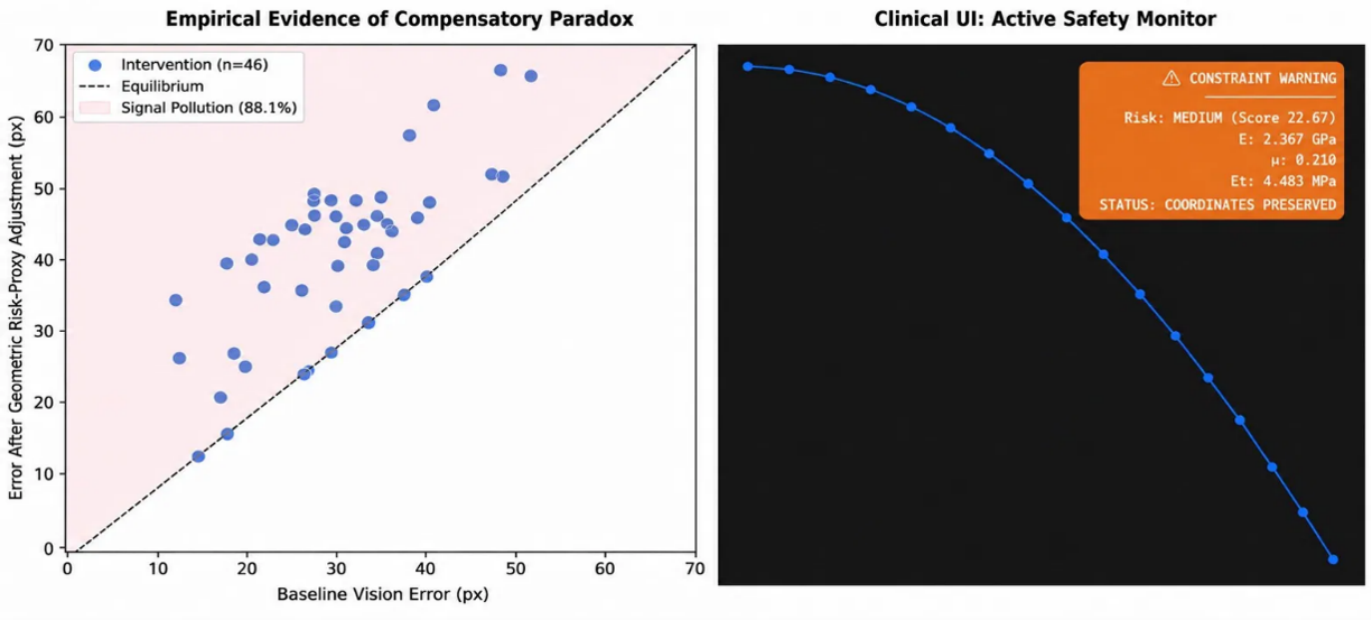


Fig. S7. **Empirical basis of the warning-only architecture.** The left panel compares baseline visual error with the error observed after geometry-derived risk-proxy intervention, and highlights the signal-pollution region in which low-resolution physical estimates overwrite high-resolution pixel structure. The right panel illustrates the active safety-monitor interface, which reports risk while preserving the original visual coordinates. Across the physics-triggered interventions evaluated, Sobel-derived contour constraints improved localization in 21.4% of cases while YOLO-based constraints improved 0.0%, and 88.1% of physically initiated coordinate modifications either failed to attenuate geometric error or introduced tracking instability.

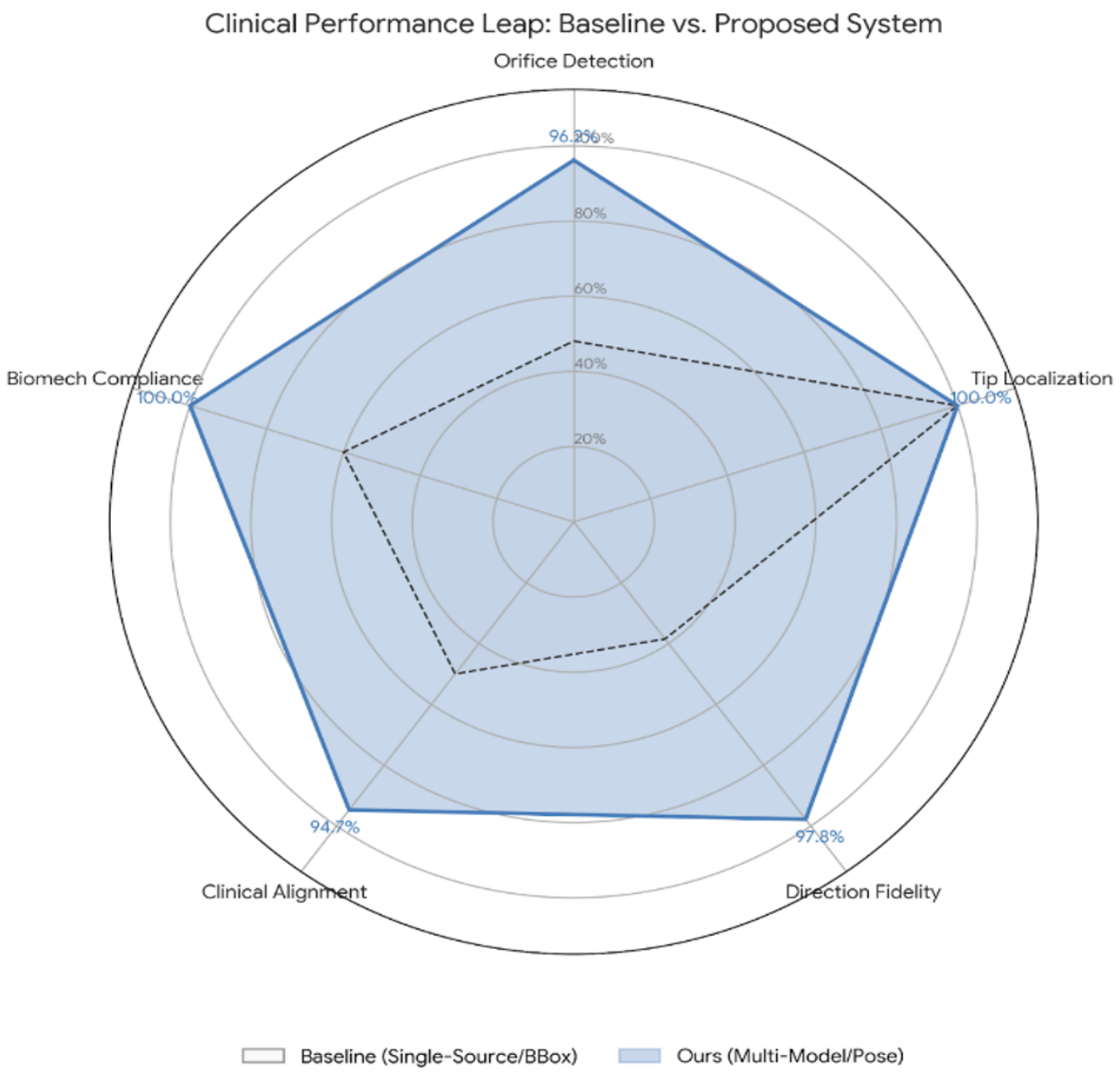


Fig. S8. **Multidimensional performance envelope.** Balance of the physio-visual architecture across accuracy, robustness, and safety. The envelope is a summary visualization of the evidence reported in the main text and in this Supplementary Material; it introduces no additional measurements. The radar label “Orifice Detection” denotes the reported navigation-target coordinate availability after cascade fusion (50/52, 96.2%), not detection of the true papillary opening. “Biomech Compliance” denotes reference-interval inclusion of geometry-derived proxies, not measured tissue properties or clinical safety.

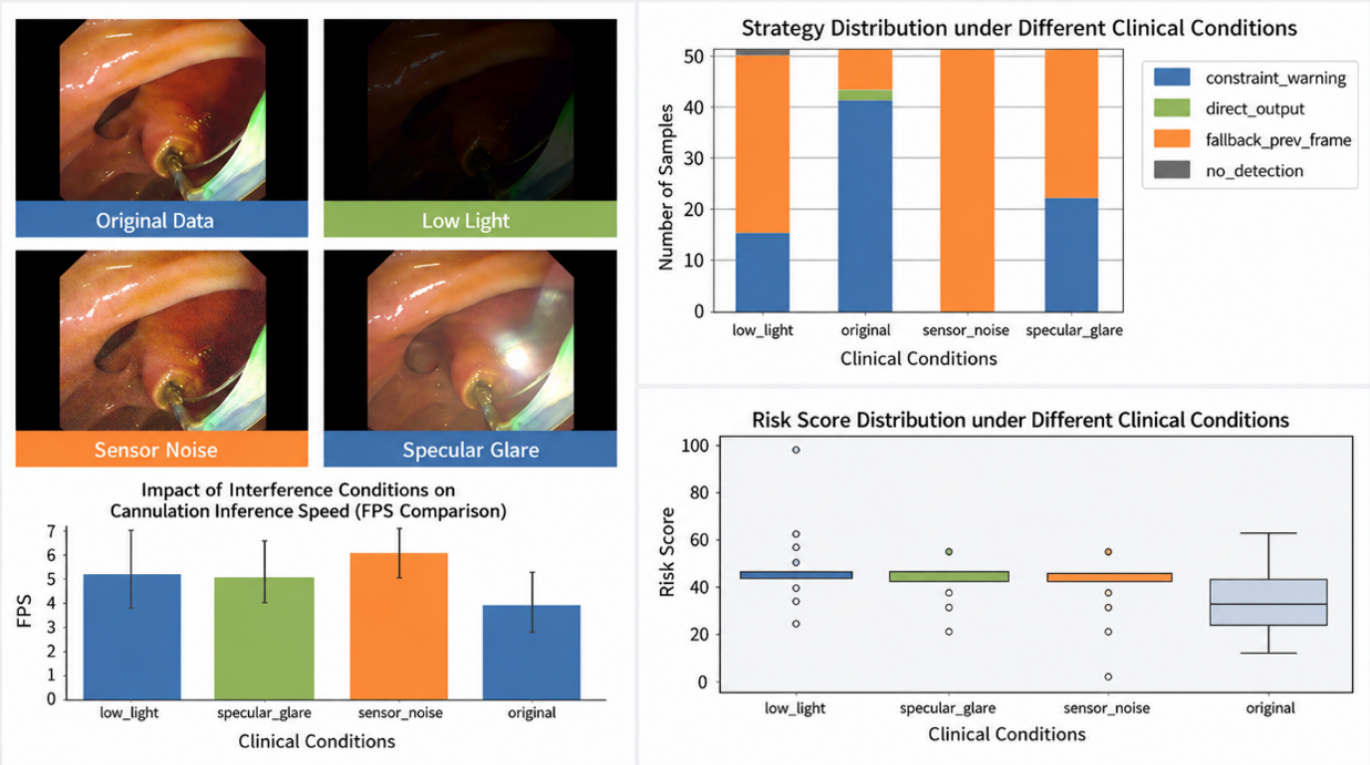


Fig. S9. **Stress-testing protocol.** System robustness under common visual disturbances including glare interference and endoscope jitter. Disturbance conditions were applied retrospectively to archived frames; the protocol is an engineering stress test and does not reproduce prospectively collected extreme conditions.

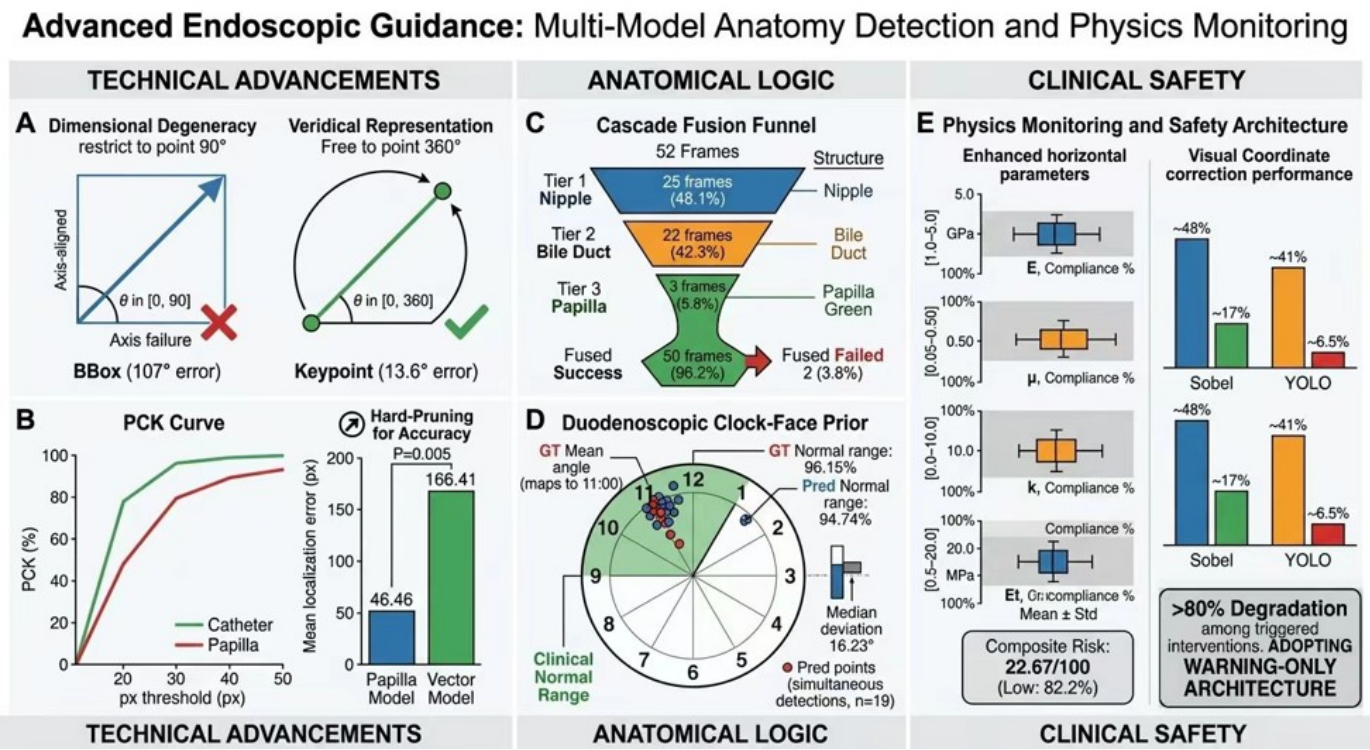


Fig. S10. **Unified view of the five experiments.** Integration of the core findings supporting the complete design logic of the system: (A) remediation of geometric degeneracy by keypoint representation; (B) localization performance and the hard-pruning benefit of the three-model architecture; (C) stepwise improvement in anatomical coverage conferred by the three-tier cascade; (D) internalization of the 11-o'clock anatomical prior; and (E) reference-interval inclusion of the risk-index proxies together with the empirical basis for the warning-only architecture.

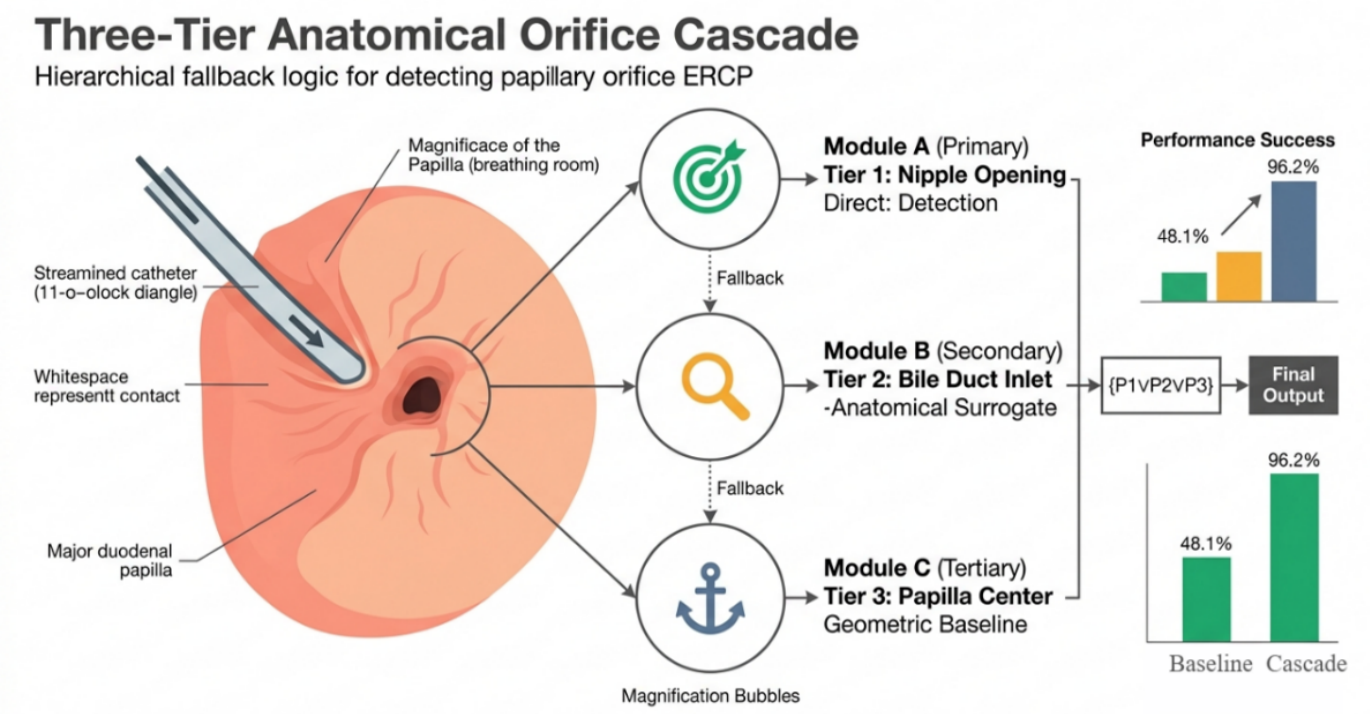


Fig. S11. **Three-tier anatomical orifice cascade.** Hierarchical fallback from nipple-opening detection to the bile-duct-inlet anatomical surrogate and the papilla-center geometric baseline, increasing navigation-target coordinate availability from 48.1% to 90.4% and finally 96.2%. The first valid detection terminates the cascade for that frame; no coordinate averaging or score-weighted fusion is performed.

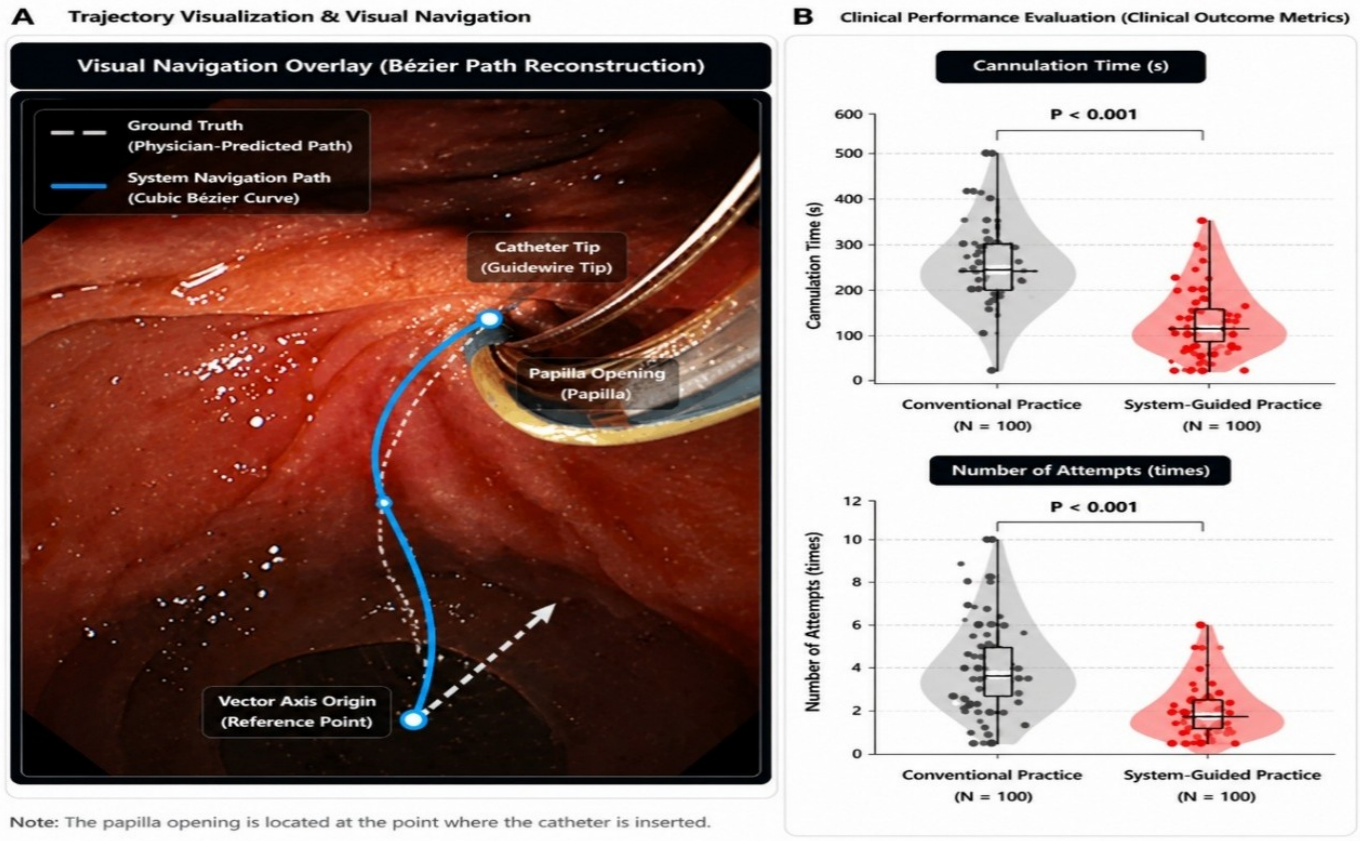


Fig. S12. **Trajectory visualization and prospective clinical-evaluation concept.** (A) Example of the cubic Bézier navigation overlay linking the vector-axis origin, the catheter tip, and the papillary opening. The overlay is a display reference only and is never fed back into coordinate control. (B) Proposed outcome displays for a future prospective comparison of conventional and system-guided practice; these illustrative distributions are not results of the present retrospective validation study.



Fig. S13. **Clinical operational interface.** Overall layout of the warning-only hybrid architecture. The left visual-stream panel displays the orifice cascade status and catheter depth index in real time; the central overlay reconstructs the predicted cannulation path as a Bézier curve and annotates the current frame rate and anatomical detection status; the right physical-stream panel outputs the four biomechanical parameters derived by the random forest regressor together with the composite risk score. The three panels correspond within a single interface to the three sequential stages of the architecture, enabling the operating clinician to access visual localization, physical compliance, and safety-alert information without leaving the endoscopic visual field.

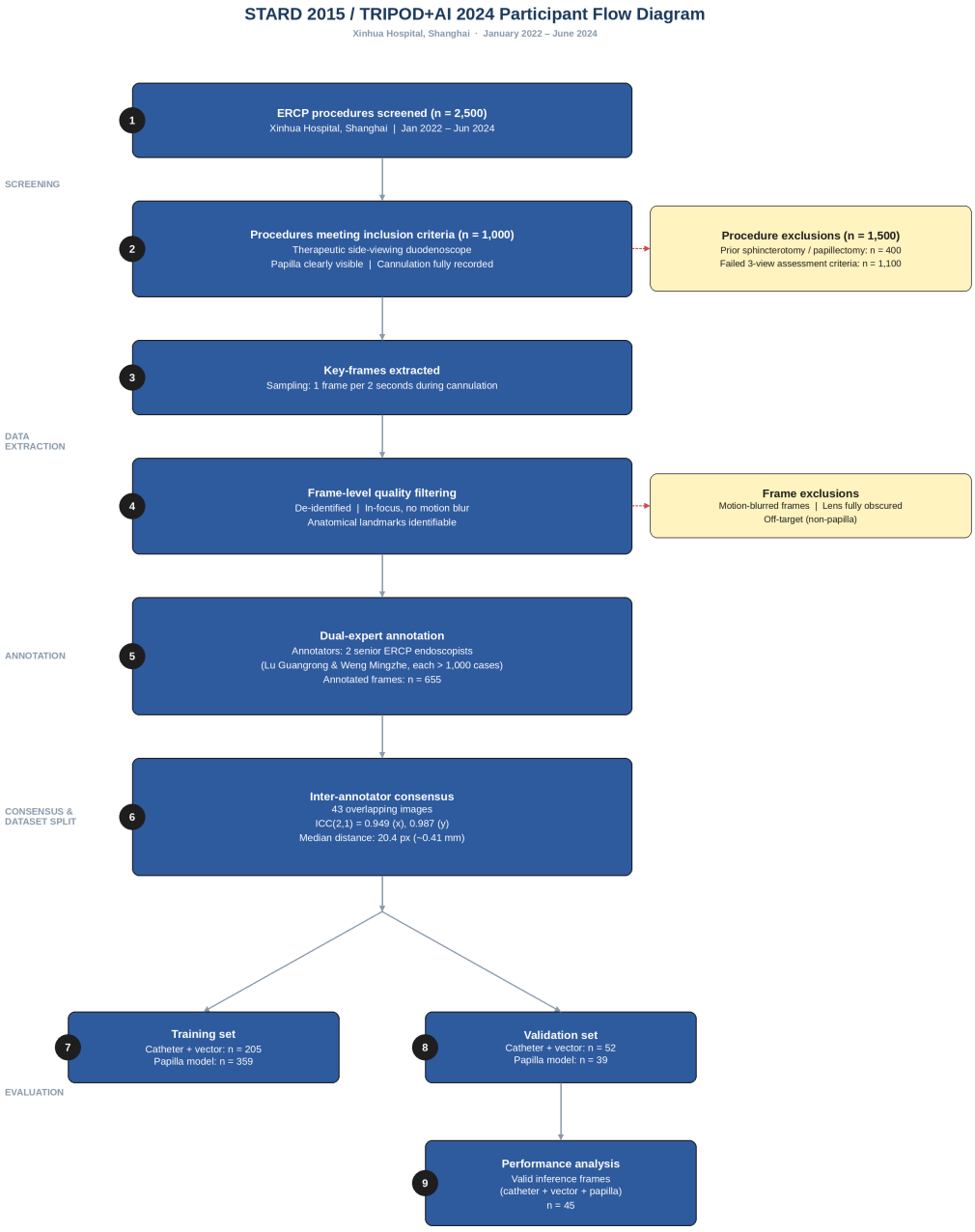


Fig. S14. **Participant screening and data flow.** Flowchart from initial procedure screening to final evaluation cohort assembly, showing the transition from the retrospective video source through eligibility screening and quality control to the 257 catheter/vector annotation frames and 398 papillary landmark annotation frames, and thence to the task-specific training and held-out validation partitions reported in the main text.

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